

## Chapter: Light – Reflection and Refraction

1. Draw ray diagrams showing the image formation by a concave mirror when an object is placed
  - a. between the pole and the focus of the mirror
  - b. between the focus and the centre of curvature of the mirror
  - c. at the centre of curvature of the mirror
  - d. a little beyond the centre of curvature of the mirror
  - e. at infinity
2. Draw ray diagrams showing the image formation by a convex lens when an object is placed
  - a. between the optical centre and the focus of the lens
  - b. between the focus and twice the focal length of the lens
  - c. at twice the focal length of the lens
  - d. at infinity
  - e. at the focus of the lens
3. Write the laws of refraction. Explain the same with the help of a ray diagram, when a ray of light passes through a rectangular glass slab.
4. Draw ray diagrams showing the image formation by a concave lens when an object is placed
  - a. at the focus of the lens
  - b. between the focus and twice the focal length of the lens
  - c. beyond twice the focal length of the lens
5. Draw ray diagrams showing the image formation by a convex mirror when an object is placed
  - a. at infinity
  - b. at a finite distance from the mirror
6. The image of a candle flame formed by a lens is projected onto a screen on the other side. If the image is three times the size of the flame and the distance between the lens and the image is 80 cm, at what distance should the candle be placed from the lens? What is the nature of the image at a distance of 80 cm and the lens?
7. The size of the image of an object by a mirror having a focal length of 20 cm is observed to be reduced to  $\frac{1}{3}$ rd of its size. At what distance has the object been placed from the mirror? What is the nature of the image and the mirror?
8. Define the power of a lens. What is its unit? One student uses a lens with a focal length of 50 cm, and another uses a lens with a focal length of  $-50$  cm. What is the nature of the lens and its power used by each of them?
9. A student focused the image of a candle flame on a white screen using a convex lens. He noted down the position of the candle screen and the lens as follows:

Position of candle = 12.0 cm  
Position of convex lens = 50.0 cm  
Position of the screen = 88.0 cm

  - (i) What is the focal length of the convex lens?
  - (ii) Where will the image be formed if he shifts the candle towards the lens at a position of 31.0 cm?

- (iii) What will be the nature of the image formed if he further shifts the candle towards the lens?
- (iv) Draw a ray diagram to show the formation of the image in case (iii) as said above.

## Chapter: Electricity

1. Three 100 W incandescent bulbs are connected in series in an electric circuit. In another circuit, another set of three bulbs of the same wattage is connected in parallel to the same source.
  - a. Will the bulb in both circuits glow to the same brightness? Justify your answer.
  - b. Now, let one bulb in both circuits get fused. Will the rest of the bulbs continue to glow in each circuit? Give a reason.
2. State Ohm's law? How can it be verified experimentally? Does it hold good under all conditions? Comment.
3. What is the electrical resistivity of a material? What is its unit? Describe an experiment to study the factors on which the resistance of a conducting wire depends.
4. How will you infer with the help of an experiment that the same current flows through every part of the circuit containing three resistances in series connected to a battery?
5. How will you conclude that the same potential difference (voltage) exists across three resistors connected in a parallel arrangement to a battery?
6. What is Joule's heating effect? How can it be demonstrated experimentally? List its four applications in daily life.

## Chapter: Magnetic Effects of Electric Current

1. Why does a magnetic compass needle pointing North and South in the absence of a nearby magnet get deflected when a bar magnet or a current-carrying loop is brought near it? Describe some salient features of the magnetic lines of field concept.
2. With the help of a labelled circuit diagram, illustrate the pattern of field lines of the magnetic field around a straight, long, conducting wire carrying current. How is the right-hand thumb rule useful to find the direction of the magnetic field associated with a current-carrying conductor?
3. Explain with the help of a labelled diagram the distribution of the magnetic field due to a current through a circular loop. Why is it that if a current-carrying coil has  $n$  turns, the field produced at any point is  $n$  times as large as that produced by a single turn?
4. Describe the activity that shows that a current-carrying conductor experiences a force perpendicular to its length and the external magnetic field. How does Fleming's left-hand rule help us to find the direction of the force acting on the current-carrying conductor?

5. Draw a labelled circuit diagram of a simple electric motor and explain its working. In what way are these simple electric motors different from commercial motors?
6. Explain the phenomenon of electromagnetic induction. Describe an experiment to show that a current is set up in a closed loop when an external magnetic field passes through the loop.
7. Describe the working of an AC generator with the help of a labelled circuit diagram. What changes must be made in the arrangement to convert it to a DC generator?
8. Draw an appropriate schematic diagram showing common domestic circuits and discuss the importance of a fuse. Why is it that another fuse of identical rating should replace a burnt-out fuse?